

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q14-0334

Murdock 230 kV  
January 01, 2016

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0334. The Interconnection Customer proposes a 168.2 MW natural gas combined cycle facility interconnecting at the Murdock 230 kV substation in the Indiana area. The request was received on May 16, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 180 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,432,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

Swan Vly 500-230 kV Autotransformer: 101.5% loading (1,041 MVA) following the outage of the Swanvly / Trego 500 KV; DFAX = 15.2%.

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                            | Required Upgrade                           | Cost Estimate | Allocation to Project |
|-------------------------------------|--------------------------------------------|---------------|-----------------------|
| Swan Vly 500-230 kV Autotransformer | Replace transformer with higher-rated unit | \$24,499,000  | 6% (\$1,432,000)      |
|                                     | Total allocated to project                 |               | \$1,432,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.